The Agony and the Ecstasy

Constructing a "Crash-Filtered" Equity Index using Machine Learning

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The Agony and Ecstasy

Passive indices are exposed to both extremes

Long-run index returns are driven by a few high-performing stocks, while a considerable share of constituents suffer catastrophic declines.

Ecstasy

- Extreme winners carry a large share of long-run wealth creation.
- Winners can be volatile, expensive, or distressed.
- Thus, even a small amount of disregarded winners can hurt performance.

Agony

- Some constituents experience deep and persistent drawdowns without recovery.
- Passive strategies may hold imploding firms for too long.

Solution

The challenge is not to eliminate highly volatile stocks. It is to distinguish recoverable volatility from persistent implosion. The empirical question is whether catastrophically declining firms are forecastable using ideas from credit-risk modelling, but with market-based distress signals.

Research Question

Main research question

To what extent does a “Crash-Filtered” equity index, constructed via probabilistic implosion modeling, generate superior risk-adjusted returns compared to the market benchmark and traditional minimum-volatility strategies?

The probabilistic implosion model follows the Catastrophic Stock Implosion approach proposed by Tewari et al. (2024). Instead of relying only on formal bankruptcy declarations, CSI uses market-based price-path signals.

Subquestions:

- How does the integration of Autoencoders for feature engineering impact the Average Precision of Ensemble models compared to those trained solely on raw financial data?
- To what extent do Ensemble methods reduce the False Positive Rate (classifying recoverable volatility as CSI) compared to traditional volatility-based exclusion strategies, while maintaining Recall?
- Which features are most important for distinguishing between "Zombie" firms (CSI) and non-zombie firms?

Dataset

Empirical design

The final panel keeps only observable CRSP firm-years: each row has the required monthly return and market-cap scaffold. Temporary and permanent CSI use the same 188,460-row scaffold; unresolved $y = \mathrm{NA}$ rows are retained in the panel but excluded from supervised model metrics.

Data sources

CRSP
monthly returns, market cap, delistings, wealth paths

Compustat
annual accounting fundamentals

FRED
macro variables

Universe

Minimum size
market cap $\geq$ USD 100M

Observable scaffold
188,460 annual firm-year rows in both tracks

Index buckets
Total, Large, Mid, Small Cap

Sample period

Coverage
Jan 1993 – Dec 2024 (32y, monthly frequency)

Supervised labels
train/test labelled; OOS may retain unresolved labels

metrics use labelled rows only

Temporal split

Train/CV	Test	OOS
143,173 rows	18,111 rows	27,176 rows

1993 2015 2019 2024

Methodology I: Response Variable Classification

Following Tewari et al. (2024), a stock is classified as a CSI candidate if it satisfies all three path criteria:

- **Initial Crash (C):** more than a C drawdown from the trailing peak marks the beginning of the "zombie" period. $C \leq -80\%$
- **Non-Recovery (M):** maximum cumulative return remains below the recovery ceiling M during the zombie period. $M \leq -20\%$
- **Zombie Period (T):** duration of the confirmation window during which non-recovery must hold. $T = 18$ months

The prediction task is annual. A trigger in calendar year $t+1$ gives the firm-year observation at t a positive label when the track-specific confirmation rule is met:

$$y_{i,t} = \begin{cases} 1, & \text{if stock } i \text{ triggers a qualifying CSI event within } [t, t+h], \\ 0, & \text{otherwise.} \end{cases}$$

Applying the paper classification

By strictly applying the methodology proposed by **?**, one finds 8,517 response observations for the selected CRSP-dataset within the 1993 to 2024 time period. The prevalence of the response variable is similar across the Train and Test datasets. Overall, the whole dataset has 188,460 observations, with roughly 75% being clustered in the Train set.

Split	Obs. rows	$y = 0$	$y = 1$	Labelled	Prev.
Full	188,460	171,269	**8,517**	179,786	4.74%
Train	143,173	136,630	6,543	143,173	4.57%
Test	18,111	17,307	804	18,111	4.44%
OOS	27,176	17,332	1,170	18,502	6.32%

Motivation for robustness

Even though the paper outlines that a combination of classification grid-values were tested, the selected combination still appears arbitary. For this reason, further robustness checks will be conducted to evaluate the chosen grid-values. It will be asked whether the dataset actually contains the bankrupt firms and how different the prevalence-rate of the response variable is when testing different grid-combinations.

Robustness I: Do CSI-firms go bankrupt?

Validation question: Among firms that eventually delist or bankrupt, how many had already been classified as CSI?

Scenario	CSI rows	CSI firms	CRSP 572-574 firms	Detected	Missed	Detection rate
Before: `confirmed_csi` only	8,369	5,642	629	151	478	24.0%

CRSP bankruptcy-related delisting codes 572, 573, and 574 are used as the terminal-distress audit set.

Core issue: The low-detection rate causes methodological issues. If one sticks strictly to the paper, then the CSI methodology should catch firms which do not recover anymore. Yet, the data shows a different result.

Methodology II: "Temporary-CSI"

Revisions to the classification methodology

The original CSI-classification grid remains unchanged. It is proposed to explicitly add all firms, which match the CRSP-codes 572,573 and 574, to the dataset after observing a valid drawdown trigger. This classification methodology will henceforth be referred to as "Temporary-CSI".

Drawdown trigger $D_{i,m} \leq C$ → Confirmation window T months → $y = 1$ CSI no recovery to M

Drawdown trigger $D_{i,m} \leq C$ → Confirmation window $\mathit{trigger} < \mathit{dlstdt} \leq \mathit{trigger} + T$ → $y = 1$ TERMINAL FAILURE $\mathit{dlstcd} \in \{572, 573, 574\}$

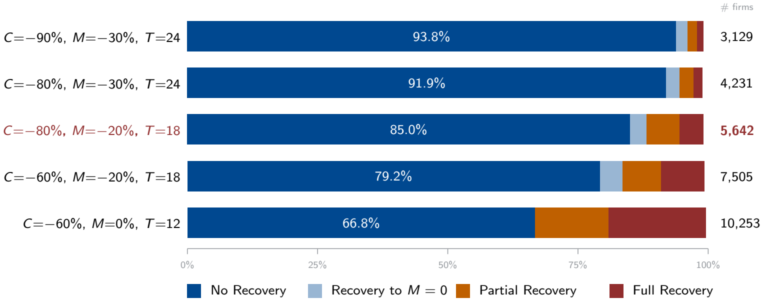
Scenario	Bankrupting	Detected	Missed	**Detection**
Before: `CSI` only	629	151	478	24.0%
After: `CSI` + terminal failure	629	**545**	84	**86.65%**

The 84 remaining misses are treated as methodological exclusions under the retained CSI-trigger rule, not as known pipeline errors.

Robustness II: Do CSI-classified firms recover again?

In addition to bankruptcy checks, robustness checks were also conducted to determine whether CSI-classified firms are in persistent decline.

A grid-search for the three classification paramaters was conducted to track the five-year recovery of firms after classification. Subsequently, each firm is sorted into one of five buckets depending on its five-year performance after classification.

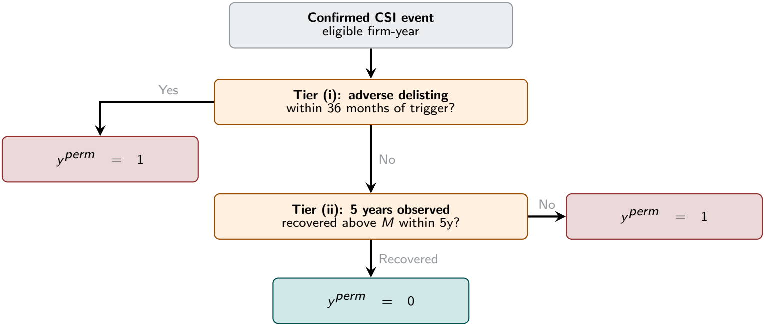


Buckets classify each CSI firm by itsfive-year post-classification performance. Light blue = the firm performs below the defined M threshold. Orange = the firm exceeds the M threshold, but does not recover entirely from the drawdown. Red = the firm recovers entirely from the drawdown.

Methodology III: “Permanent-CSI”

Dealing with recovering firms

To deal with the prior outlined false-positives due to observing a non-neglectible portion of recovering firms, it is proposed to test a secondary classification methodology: ”Permanent-CSI”. This methodology is even further forward looking, if a firm did not recover above the defined M threshold after 5 years, it will be classified as a CSI-event.



Applying the classification: Permanent CSI

By construction, Permanent-CSI is more restrictive than the initial classification methodology. The whole datasets now records 6,258 responses with y = 1 (compared to 8,517 originally). The prevalence drops from 4,74 to 3,34%. Overall, the prevalence rate is similar amongst the train and test set, with the test set showing a slightly smaller prevalence of the response variable.

Split	Obs. rows	$y = 0$	$y = 1$	Labelled	Prev.
Full	188,460	181,368	**6,258**	187,626	3.34%
Train	143,173	137,794	5,379	143,173	3.76%
Test	18,111	17,511	542	18,053	3.00%
OOS	27,176	26,063	337	26,400	1.28%

Low prevalence in the OOS set

The OOS dataset shows a markedly smaller prevalence rate of the response variable. The reason being that the forward-looking classification-methodology is constrained by the available data periods. Yet, this is not an issue as one does not require labelling the OOS dataset, which will be used to mainly evaluate the index-construction performance at the later part of the thesis.

Dataset: Descriptive Statistics

CSI positives are rare and economically distinct from non-positives

Both label definitions isolate a small distressed tail. The key contrast is not only prevalence: positive firms are much smaller and show worse realized returns and profitability.

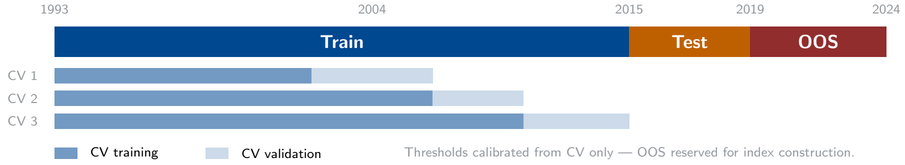
Sample	CSI	y	Obs.	Share	Firms	Median mcap	Ann. ret.	ROA
Full	Temp.	0	171,269	95.26%	19,073	254.9	4.9%	1.6%
Full	Temp.	1	**8,517**	4.74%	5,603	**74.9**	**-35.0%**	**-18.8%**
Full	Perm.	0	181,368	96.66%	19,565	252.6	4.0%	1.5%
Full	Perm.	1	6,258	3.34%	4,696	**65.7**	**-37.5%**	**-20.0%**

Remark: Market cap is USD millions.

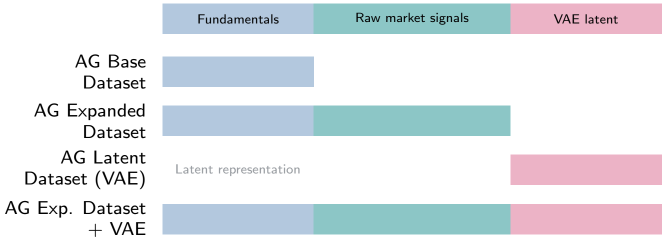
Even though the classification methodology seems to seperate firms rather well, it also constrains the expectations for the index-construction part of this thesis. CSI-classified firms are considerably smaller (measured by market cap.) than their peers. Thus, their index weight will be smaller, which implies only a minor minor contribution to overall index-returns. Consequently, a priori one would not expect to observe a drastic outperformance by avoiding or downweighting CSI-classified firms.

Modelling I: Setup and Feature Engineering

Validation design



Final feature-set keys



Modelling II: Results Temporary CSI

Description

Temporary CSI is a rare-event classification problem: only 4.74% of labelled firm-years are positives. Because of this class imbalance, AP and fixed-FPR recall are more decision-relevant than AUC alone. AG Exp. Dataset + VAE ranks best in CV, while AG Expanded Dataset is strongest on the 2016-2019 test split.

Model	CV-AP	CV-AUC	CV-FPR3	Test-AP	Test-AUC	Test-FPR3
AG Expanded Dataset	0.2046	0.8635	0.2344	**0.1985**	**0.8766**	**0.2002**
AG Base Dataset	0.1923	0.8446	0.2206	0.1656	0.8538	0.1517
AG Latent Dataset (VAE)	0.1659	0.8454	0.1706	0.1501	0.8438	0.1294
AG Exp. Dataset + VAE	**0.2114**	**0.8667**	**0.2478**	0.1864	0.8741	0.1779

Remark: CV-FPR3 and Test-FPR3 are R@FPR3. OOS, R@FPR1, R@FPR5, Brier, and full split tables are in Appendix A10.

Interpretation

Test-AP improves from the 4.44% test prevalence baseline to 19.85%, showing meaningful concentration of CSI cases near the top of the risk ranking. Test-AUC of 0.8766 confirms strong broad ranking, but AP is the more useful indicator for practical screening.

Modelling III: Results Permanent-CSI

Description

Permanent CSI is the harder task: positives are rarer than temporary CSI, with only 3.34% positives in the full labelled sample and 3% in the test split. Despite this, the models still rank risk well. AG Expanded Dataset remains the reporting baseline, while AG Exp. Dataset + VAE is the strongest non-raw challenger on CV/test.

Model	CV-AP	CV-AUC	CV-FPR3	Test-AP	Test-AUC	Test-FPR3
AG Expanded Dataset	0.1854	**0.8757**	**0.2607**	**0.1416**	0.8810	0.1808
AG Base Dataset	0.1785	0.8677	0.2379	0.1289	0.8627	0.1697
AG Latent Dataset (VAE)	0.1426	0.8460	0.1960	0.1115	0.8496	0.1513
AG Exp. Dataset + VAE	**0.1883**	0.8719	0.2578	0.1415	**0.8838**	**0.2011**

Remark: CV-FPR3 and Test-FPR3 are R@FPR3. OOS, R@FPR1, R@FPR5, Brier, and full split tables are in Appendix A11.

Interpretation

Temporary CSI has more positives and higher test prevalence: 4.44% vs 3%. Permanent CSI is therefore harder and more affected by class imbalance. AUC is similar or slightly higher for permanent CSI, but AP is lower because the base prevalence is lower. AG Expanded Dataset and AG Exp. Dataset + VAE are nearly tied on Test-AP: 0.1416 vs 0.1415. AG Exp. Dataset + VAE has the best Test-AUC and Test-FPR3, so it is a strong challenger, but not a clean replacement for the reporting baseline.

Modelling IV: What VAE Features Add

Feature-set interpretation

The VAE does not replace expanded-dataset predictive information. It is useful as an augmentation or robustness signal: AG Exp. Dataset + VAE improves temporary OOS AP/FPR recall, while AG Latent Dataset (VAE) is the strongest permanent OOS model.

Temporary CSI

- AG Exp. Dataset + VAE leads OOS AP: 0.3152.
- OOS R@FPR1/3/5 improves to 0.1051 / 0.2632 / 0.4128.
- AG Expanded Dataset still has the slightly higher OOS AUC: 0.8961 vs 0.8950.

Permanent CSI

- AG Exp. Dataset + VAE leads test AUC and FPR3/5 recall.
- AG Latent Dataset (VAE) leads OOS AP/AUC/Brier/FPR recall.
- AG Base Dataset remains a useful fundamentals-only benchmark, not the preferred model.

Model-family note

Weighted AutoGluon ensembles top the leaderboard across tracks and feature sets. Tree families supply most of the useful base-model signal; neural nets are considered but do not dominate the final ranks. Standalone XGBoost diagnostics exist for the main model runs, but they are not relabelled here because the final index grid has no XGB-only strategy rows.

CV-Only False-Positive Diagnostic

Training/CV cohorts only; false positives are not cleanly separable in available evidence

Track	Model	FP	TP	Near thr.	TP-score overlap	Above TP med.	TP cost /100 near FP
Temp. CSI	raw+latent	2,059	773	16.6%	72.7%	44.5%	31.6
Perm. CSI	raw+latent	2,092	634	18.6%	64.3%	37.0%	20.0
Temp. CSI	raw	2,073	731	18.1%	69.8%	46.6%	29.5
Perm. CSI	raw	2,092	641	16.7%	69.6%	41.9%	26.0

Interpretation boundary

At the CV FPR3 rule, only 16.6–18.6% of false positives are narrowly threshold-adjacent. Most overlap the true-positive score range, and raising thresholds would also remove true positives. Row-level raw/fundamental/latent feature matrices keyed by firm-year were not available, so feature-space separability and mechanism labels remain unclaimed.

Remark: CV-only. Sources: `AE-FP-DIAG-005_presentation_summary_table.csv`; `AE-FP-DIAG-005_Presentation_Ready_CV_FP_Summary.md`. No test/OOS labels, outcomes, rows, or features are used.

Index Construction I: From Score to Portfolio Weight

One score rule, two exclusion regimes

The model produces a probability of CSI next year. A threshold turns that probability into a binary exclusion flag. The flag is then applied to market-cap weights. Two regimes differ only in how long the flag persists after it first fires.

From model score to portfolio weight

Score
probability of CSI next year
from AutoGluon

→

Exclusion flag
1 if score $\geq$ threshold
FPR 1%, 3%, 5%, Youden J

→

Portfolio weight
market-cap $\times$ $(1 - \text{flag})$
renormalized

How the flag evolves over time — a firm crosses the threshold in year 3

Temporary CSI (lockout $L = 3$)	y1	y2	y3 $\star$	y4	y5	y6	y7
Permanent CSI (absorbing)	y1	y2	y3 $\star$	y4	y5	y6	y7

Held | First flag ($\star$) | Excluded

Index Construction II: Four Benchmark Universes

Same exclusion rule, four universes

The revised 11C track applies the score-to-weight pipeline independently inside each CRSP-like universe, producing four screened benchmarks alongside their unfiltered counterparts.

Total Market	**Large Cap**	**Mid Cap**	**Small Cap**
full universe $\geq$ USD 100M	CRSP-like top tier	CRSP-like middle tier	CRSP-like bottom tier

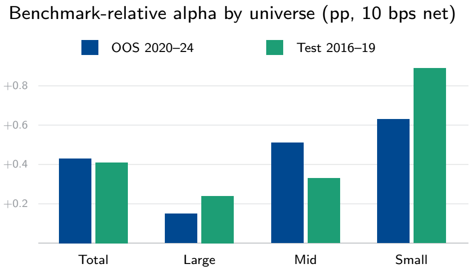
Benchmark construction

- Universe formed from prior-December market capitalization.
- Minimum market cap: USD 100 million.
- Quarterly rebalancing: March, June, September, December.
- Market-cap weights; cost overlays use 0, 5, 10, and 20 bps.

Signal discipline

- Scores are annual for AG Expanded Dataset, AG Base Dataset, AG Latent Dataset (VAE), and AG Exp. Dataset + VAE.
- Holding-year signal uses information available before the holding year.
- Thresholds calibrated from CV predictions only: Youden, FPR1, FPR3, FPR5.

Index Results: Does the Crash Filter Help?



OOS alpha. Current OOS deltas are positive in all four universes; Mid and Small are strongest.

Test-set check. Test evidence is shown separately; Permanent-CSI Total Sharpe rises $0.89 \rightarrow \mathbf{1.00}$.

Source of alpha. Mostly retained-name reweighting, not direct event-avoidance (next slides).

Summary

The crash-filter appears beneficial in both evaluation windows, but the evidence differs by split. The OOS result is a positive benchmark-relative return delta across the four universes. The test-set result is reported separately and should not be read as the same diagnostic as OOS risk performance.

Temporary-CSI Results: OOS (2020-2024) with 10 bps of Transaction Costs

Best strategy per universe vs. the market-weighted benchmark assuming 10 bps of transaction costs.

		Return		Risk	
Universe	Strategy	Geo ret.	Δ pp	Sharpe	Max DD
Total	AG Base Dataset; Youden 3y	13.3 → 13.7	+0.43	0.60 → **0.64**	−25.0 → **−23.8**
Large	AG Base Dataset; Youden 3y	14.3 → 14.4	+0.15	0.66 → **0.68**	−24.8 → **−23.9**
Mid	AG Base Dataset; FPR5 5y	9.5 → 10.0	+0.51	0.41 → **0.42**	−24.6 → **−24.5**
Small	AG Exp. Dataset + VAE; Youden 3y	7.6 → 8.2	**+0.63**	0.31 → **0.33**	−29.1 → −29.3

Main takeaway

The risk-filter works better the smaller the market cap.: the alpha rises from +0.15pp (Large) to +0.51pp (Mid) to +0.63pp (Small). Smaller-cap universes include more firms in distress and exclusions move a larger part of the portfolio (about 85-106% turnover vs. 10-13% for the total market). Drawdown improves everywhere except for small caps.

Permanent-CSI Results: OOS (2020-2024) with 10 bps of Transaction Costs

Best strategy per universe vs. the market-weighted benchmark assuming 10 bps of transaction costs.

		Return		Risk	
Universe	Strategy	Geo ret.	Δ pp	Sharpe	Max DD
Total	AG Exp. Dataset + VAE; FPR5	13.3 → 13.5	+0.27	0.60 → **0.61**	−25.0 → **−24.7**
Large	AG Exp. Dataset + VAE; FPR5	14.3 → 14.5	+0.23	0.66 → **0.67**	−24.8 → **−24.5**
Mid	AG Base Dataset; FPR5	9.5 → 10.3	**+0.74**	0.41 → **0.44**	−24.6 → **−24.5**
Small	AG Latent Dataset (VAE); FPR3	7.6 → 7.9	+0.32	0.31 → **0.32**	−29.1 → −29.3

Main takeaway

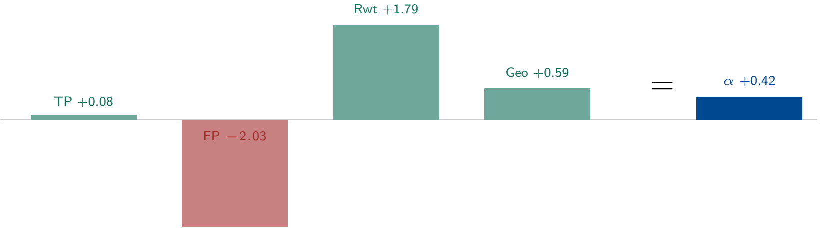
The permanent filter tilts the gains toward Mid Cap (+0.74pp), its strongest universe. Direction matches the temporary track, but more exclusion over-removes firms with broad thresholds, which is why Youden is dropped in favour of FPR3 and FPR5.

Where Does the Alpha Come From?

It is not direct event-avoidance

Because positives are rare, realized alpha is dominated by reweighting retained firms, not by directly dodging CSI events. A large false-positive opportunity cost is the main offset. For this reason, most of the performance gain disappears.

Decomposition for Total Market, OOS (pp): TP + FP + reweight + cost + geo = alpha



Universe (OOS, temp)	TP	FP	Reweight	TC	Geo	Alpha
Total	+0.08	−2.03	+1.79	−0.01	+0.59	+**0.42**
Large	+0.00	−1.50	+1.24	−0.01	+0.41	+**0.14**
Mid	−0.00	+0.05	+0.27	−0.11	+0.18	+**0.39**
Small	+0.42	−4.00	+3.94	−0.09	+0.28	+**0.55**

Permanent-CSI Results: Test (2016-2019) with 10 bps of Transaction Costs

		Return		Risk	
Universe	Strategy	Geo ret.	Δ pp	Sharpe	Max DD
Total	AG Exp. Dataset + VAE; Youden	13.8 → 14.2	+0.45	0.89 → **1.00**	−14.6 → **−12.8**
Large	AG Exp. Dataset + VAE; Youden	14.2 → 14.4	+0.26	0.95 → **1.04**	−13.8 → **−12.2**
Mid	AG Exp. Dataset + VAE; Youden	13.7 → 14.2	+0.43	0.84 → **0.95**	−15.3 → **−13.8**
Small	AG Exp. Dataset + VAE; Youden	11.4 → 12.2	**+0.83**	0.58 → **0.68**	−20.1 → **−18.3**

Validation Summary

In the calmer 2016-2019 window a single consistent strategy (Exp. Dataset + VAE with Youden computed thresholds) improves Sharpe and cuts drawdown in every universe, with Total Sharpe crossing 1.0. The same small-cap tilt appears (+0.83pp Small vs. +0.26pp Large).

Impact of transaction costs: the winners are unchanged

Selected OOS winners when assuming transaction costs of 5, 10 and 20 bps.

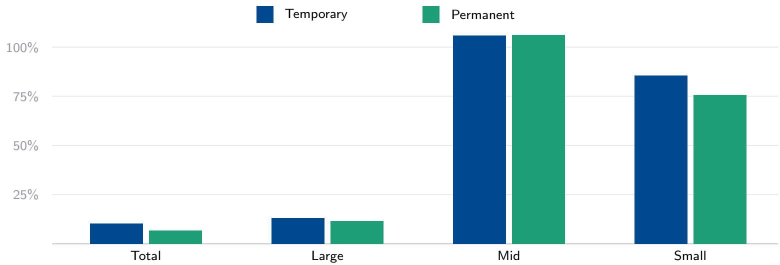
Track	Universe	Selected OOS winner	Active α (5/10/20)	Ranking
Temp.	Total	AG Base Dataset; Youden 3y	$+0.43/+0.43/+0.42$	unchanged
Temp.	Large	AG Base Dataset; Youden 3y	$+0.15/+0.15/+0.15$	unchanged
Temp.	Mid	AG Base Dataset; FPR5 5y	$+0.51/+0.51/+0.51$	unchanged
Temp.	Small	AG Exp. Dataset + VAE; Youden 3y	$+0.63/+0.63/+0.62$	unchanged
Perm.	Total	AG Exp. Dataset + VAE; FPR5	$+0.27/+0.27/+0.27$	unchanged
Perm.	Large	AG Exp. Dataset + VAE; FPR5	$+0.23/+0.23/+0.23$	unchanged
Perm.	Mid	AG Base Dataset; FPR5	$+0.74/+0.74/+0.74$	unchanged
Perm.	Small	AG Latent Dataset (VAE); FPR3	$+0.32/+0.32/+0.32$	unchanged

Takeaway

Raising costs to 20 bps lowers net returns but does not change a single winner or flip the strategy ranking.

Cost Drag due to turnover is concentrated in mid and small caps

Annualized gross buy + sell turnover of the selected OOS strategies (%)



Summary

Drag scales linearly with turnover × cost. Total and Large are barely traded (between 7-13%), so their 20 bps drag is at most 0.13pp. Mid and Small rebalancing is comparably larger at 75-106% a year, lifting their 20 bps drag to roughly 0.76-1.06pp. Yet, the winner ranking on the previous slide still holds.

The Selected Rules Are Principled, Not Grid-Mined

Temporary CSI — finite lockout

Youden 3yr wins Total, Large, and Small; FPR5 5yr wins Mid. A recoverable 3–5yr lockout lets the signal age out, capping turnover outside Mid / Small.

20 bps evidence: Youden has the strongest mean best alpha, +0.35pp.

Permanent CSI — strict FPR

FPR5 wins Total, Large, and Mid; FPR3 wins Small. Absorbing removal needs narrower gates — broad Youden over-removes names and compounds turnover.

20 bps evidence: FPR3 / FPR5 lead (+0.35 / +0.33pp); Youden averages −0.20pp.

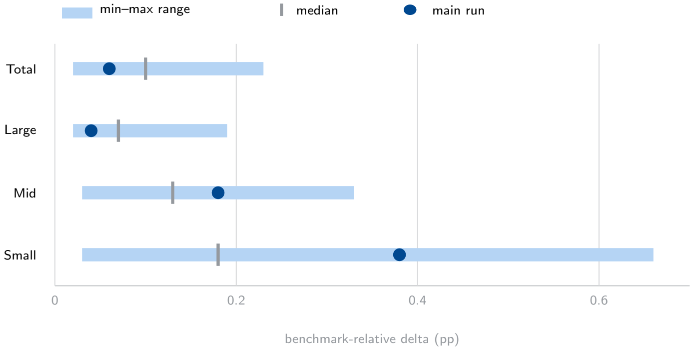
Why this matters

The two tracks select different rule families for a structural reason, not by grid-mining: a temporary flag can expire, so it tolerates a looser cutoff; a permanent flag is irreversible, so cost and turnover mistakes must be gated more tightly. Turnover detail is on the previous slide.

Remark: Source: `threshold_family_summary_20bps.csv`; `best_by_track_index_cost.csv`.

Sensitivity Analysis: temporary CSI only

Main run versus 24 completed temporary-CSI C/M/T sensitivity runs; relative geometric-return delta (pp).



Robustness read

All completed temporary-CSI C/M/T runs stay positive; sensitivity changes magnitude, not sign. Main run: below median in Total/Large, above in Mid/Small. Temporary-CSI evidence only.

Remark: Temporary CSI only; three blocked-partial configs excluded. Source: `universe_stability_summary.csv`.

Appendix A1: Delisting and Bankruptcy Detection

Indicator group	CRSP firms	Detected	Rate	Positive rows followed
Bankruptcy range 572-574	629	545	86.65%	2,757
Bankruptcy / declared insolvent 574	548	513	93.61%	2,502
Adverse 400-490, 550-585	5,654	3,676	65.02%	7,000

Why the override is additive

The CRSP signal is not an independent default label. It only turns a CSI-like trigger into a positive `dynamic_csi` event when the firm fails before the 18-month confirmation clock can fully resolve.

Appendix A2: Revised Annual Prevalence

Annual label metric	Count
Temporary positive label cells	8,517
Temporary unresolved cells ($y = NA$)	8,674
Permanent positive label cells	6,258
Permanent unresolved cells ($y = NA$)	834

Current CRSP 572–574 overlap

Metric	Count
CRSP 572–574 firms in sample	629
Detected by revised temporary CSI	545
Missed by revised temporary CSI	84
Detection rate	86.65%

The 84 are CRSP 572–574 firms not captured by a valid drawdown trigger within the terminal-failure timing window. They remain outside labelled positives, not known pipeline errors.

Methodological status

The local overlap audit supports 545 detected and 84 missed. It does not decompose every miss, so the wording stays limited to the retained trigger-and-window rule.

Appendix A3: Cleaned Label Counts

CSI type	Split	Obs. rows	$y = 0$	$y = 1$	Labelled	Prev.
Temporary	Full	188,460	171,269	**8,517**	179,786	4.74%
Temporary	Train	143,173	136,630	6,543	143,173	4.57%
Temporary	Test	18,111	17,307	804	18,111	4.44%
Temporary	OOS	27,176	17,332	1,170	18,502	6.32%
Permanent	Full	188,460	181,368	6,258	187,626	3.34%
Permanent	Train	143,173	137,794	5,379	143,173	3.76%
Permanent	Test	18,111	17,511	542	18,053	3.00%
Permanent	OOS	27,176	26,063	337	26,400	1.28%

Read

Both targets remain rare after the observable-scaffold cleanup. Unresolved labels stay in canonical artifacts for scaffold consistency, but are excluded from labelled counts shown here, supervised training, and label-based evaluation.

Source: Labelled rows $= y = 0 + y = 1$; prevalence $= y = 1$/labelled rows.

Appendix A4: Descriptive Statistics – Market Cap

Market cap, USD millions

CSI type	Split	y	Mean	Median	P25	P75
Temporary	CV	0	4,117.4	408.1	87.3	1,924.5
Temporary	CV	1	613.4	85.1	30.0	255.5
Temporary	Full	0	4,864.6	350.9	67.2	1,903.7
Temporary	Full	1	574.6	68.7	23.3	228.0
Permanent	CV	0	3,452.1	354.5	85.2	1,551.6
Permanent	CV	1	447.3	79.6	30.3	221.6
Permanent	Full	0	3,558.3	249.3	57.7	1,232.6
Permanent	Full	1	496.5	66.2	22.7	219.0

Read

CSI-positive firms are much smaller across both definitions. In the full sample, median market cap is 68.7 for temporary positives versus 350.9 for non-positives, and 66.2 for permanent positives versus 249.3 for non-positives.

Source: Market cap is USD millions.

Appendix A5: Descriptive Statistics – Other Medians

CSI type	Split	y	Ann. ret.	Assets	Sales	ROA	M/B
Temporary	CV	0	6.1%	640.9	326.7	1.7%	1.82
Temporary	CV	1	-24.9%	86.1	50.2	-14.2%	1.51
Temporary	Full	0	4.9%	565.3	295.2	1.6%	1.81
Temporary	Full	1	-35.0%	77.5	46.8	-18.8%	1.59
Permanent	CV	0	6.0%	567.8	270.2	1.7%	1.82
Permanent	CV	1	-28.3%	78.8	44.4	-15.8%	1.46
Permanent	Full	0	4.2%	395.3	191.9	1.5%	1.80
Permanent	Full	1	-37.5%	64.6	35.1	-19.3%	1.52

Read

The positive class has sharply lower realized returns, smaller operating scale, and negative profitability. The contrast strengthens in the full sample for both temporary and permanent CSI.

Source: Assets and sales are USD millions.

Appendix A6: CV Folds, Leakage Controls, and Training Metric

Expanding-window CV

Fold	Training years	Validation years
1	Initial training block	No validation; used to seed expanding window
2	Up to 1998	1999-2004
3	Up to 2004	2005-2009
4	Up to 2009	2010-2015

Training / selection metric

- **AutoGluon:** yes, model ranking uses AP via `EVAL_METRIC = "average_precision"` passed to `TabularPredictor(..., eval_metric=EVAL_METRIC)` in `09C_AutoGluon.py`.
- **XGBoost:** binary-logistic training uses `eval_metric="aucpr"` in `09_Train.R`; reported AP is computed from predicted probabilities.
- Split before preprocessing.
- Train-fitted winsorization, imputation, and PIT only.
- CV AP/AUC/R@FPR metrics are averaged per fold; OOS is reserved for index construction, not model selection.

Appendix A7: Feature Engineering Groups

Group	How it works in the final model inputs
Firm fundamentals	Point-in-time ratios turn accounting levels into leverage, liquidity, coverage, profitability, accrual, cash-flow, and Altman-style signals.
Level and momentum	First differences, second differences, lagged expanding means, peak/trough moves, consecutive-decline counters, accounting momentum, and 3/5-year rolling mean/min/max/trend/autocorrelation summarize each firm's own history.
Dispersion and price distress	Within-firm dispersion appears through lagged expanding volatility, rolling standard deviation, monthly return volatility, trailing momentum, and 12/60-month drawdown features. These are time-series distress summaries, not peer-universe cross-sectional ranks.
Macro and interactions	Macro levels and regimes enter directly, with selected firm $\times$ macro interactions such as leverage/rates, coverage/rates, debt/high-yield spreads, profitability/GDP growth, and returns/VIX.
Rank-style VAE scaling	Before VAE fitting, continuous inputs are imputed using train-fitted rules and transformed by train-fitted PIT/quantile scaling to a percentile-like $[0, 1]$ scale; test and OOS rows reuse those fitted transforms.
Feature-set assembly	**Base** strips price-direction features; **Expanded** uses the full engineered matrix; **Latent** uses VAE $z_1, \ldots, z_{24}$ plus reconstruction error; **Exp. + VAE** concatenates expanded and latent signals.
Metadata excluded	Labels, event dates, confirmation dates, censoring flags, and diagnostics are retained for audit but excluded from model features.

Appendix A8: Best-Model Hyperparameters and Choices

What won locally

- All eight track × feature-set runs rank `WeightedEnsemble_L2` first.
- Selection uses average precision, matching the rare-positive CSI setting.
- Temporary CSI: `raw_plus_latent` has the strongest CV/OOS AP case; raw remains the test/AUC comparator.
- Permanent CSI: `raw_plus_latent` is the main non-raw challenger; `latent_raw` is only an OOS robustness sensitivity.

Hyperparameter/model-choice evidence

- `WeightedEnsemble_L2`: stack level 2, greedy ensemble, `save_bag_folds = TRUE`, target ensemble size 25.
- Base menu: LightGBM variants, CatBoost, XGBoost, ExtraTrees/RandomForest, and neural nets.
- Retained `LightGBMLarge`: learning rate 0.03, 128 leaves, feature fraction 0.9, minimum leaf size 3.
- Full fitted hyperparameters are pruned for some compact evidence, so missing tuning details are not inferred.

Story

The local evidence favors nonlinear mixtures over one stable parametric form. Ensembles dominate the validation leaderboards; tree and boosted-tree families carry much of the signal, while XGBoost appears as a candidate/component rather than the final standalone winner.

Remark: Source: model-suite comparison files, family-winner CSV, compact hyperparameter JSONs, ensemble-weight JSONs, and AutoGluon log excerpts under `03_Data_Output/6_ModelSuite`.

Appendix A9: VAE Architecture Details

Component	Setting
Encoder	*n_features* $\rightarrow$ 256 $\rightarrow$ 128 $\rightarrow$ 64 $\rightarrow$ *z_mean*, *z_log_var*
Latent dimension	24
Decoder	*z* $\rightarrow$ 64 $\rightarrow$ 128 $\rightarrow$ 256 $\rightarrow$ *n_features*
Loss	Reconstruction + β KL + γ supervised CSI classification loss
Parameters	$\beta = 1.0$, $\gamma = 0.1$, early stopping patience 15
Status	Weakly supervised on training rows; test and OOS labels are not used for VAE fitting.

Appendix A10: Temporary CSI CV/Test/OOS Model Metrics

Full model-suite threshold metrics Train/CV, test, and OOS; AP/AUC/FPR recall.

Split	Model	AP	AUC	R@FPR1	R@FPR3	R@FPR5	Brier
CV	AG Expanded Dataset	0.2046	0.8635	0.0917	0.2344	0.3415	0.0371
CV	AG Base Dataset	0.1923	0.8446	0.0830	0.2206	0.3229	0.0374
CV	AG Latent Dataset (VAE)	0.1659	0.8454	0.0670	0.1706	0.2735	0.0380
CV	AG Exp. Dataset + VAE	**0.2114**	**0.8667**	**0.0981**	**0.2478**	**0.3581**	**0.0368**
Test	AG Expanded Dataset	**0.1985**	**0.8766**	**0.0821**	**0.2002**	**0.2998**	**0.0389**
Test	AG Base Dataset	0.1656	0.8538	0.0585	0.1517	0.2413	0.0392
Test	AG Latent Dataset (VAE)	0.1501	0.8438	0.0410	0.1294	0.2264	0.0394
Test	AG Exp. Dataset + VAE	0.1864	0.8741	0.0634	0.1779	0.2749	0.0394
OOS	AG Expanded Dataset	0.3084	**0.8961**	0.0915	0.2547	0.4000	0.0498
OOS	AG Base Dataset	0.2526	0.8686	0.0598	0.1795	0.3017	0.0526
OOS	AG Latent Dataset (VAE)	0.2813	0.8626	0.1009	0.2274	0.3556	0.0518
OOS	AG Exp. Dataset + VAE	**0.3152**	0.8950	**0.1051**	**0.2632**	**0.4128**	**0.0494**

Remark: Source: `03_Data_Output/6_ModelSuite/comparison/AE-MODEL-SUITE-007_model_suite_metrics_long.csv`; raw comparator from `raw/derived_metrics/raw_complete_threshold_metrics.csv`.

Appendix A11: Permanent CSI CV/Test/OOS Model Metrics

Full model-suite threshold metrics Train/CV, test, and OOS; AP/AUC/FPR recall.

Split	Model	AP	AUC	R@FPR1	R@FPR3	R@FPR5	Brier
CV	AG Expanded Dataset	0.1854	**0.8757**	0.1098	**0.2607**	**0.3697**	0.0297
CV	AG Base Dataset	0.1785	0.8677	0.1033	0.2379	0.3469	0.0299
CV	AG Latent Dataset (VAE)	0.1426	0.8460	0.0732	0.1960	0.2908	0.0308
CV	AG Exp. Dataset + VAE	**0.1883**	0.8719	**0.1196**	0.2578	0.3680	**0.0297**
Test	AG Expanded Dataset	**0.1416**	0.8810	**0.0664**	0.1808	0.3007	0.0284
Test	AG Base Dataset	0.1289	0.8627	0.0627	0.1697	0.2620	0.0293
Test	AG Latent Dataset (VAE)	0.1115	0.8496	0.0424	0.1513	0.2546	**0.0280**
Test	AG Exp. Dataset + VAE	0.1415	**0.8838**	0.0609	**0.2011**	**0.3155**	0.0283
OOS	AG Expanded Dataset	0.0323	0.8081	0.0000	0.0119	0.0623	0.0211
OOS	AG Base Dataset	0.0281	0.7798	0.0000	0.0030	0.0297	0.0232
OOS	AG Latent Dataset (VAE)	**0.0482**	**0.8337**	**0.0504**	**0.1484**	**0.2226**	**0.0166**
OOS	AG Exp. Dataset + VAE	0.0311	0.8032	0.0000	0.0208	0.0653	0.0207

Remark: Source: `03_Data_Output/6_ModelSuite/comparison/AE-MODEL-SUITE-007_model_suite_metrics_long.csv`; raw comparator from `raw/derived_metrics/raw_complete_threshold_metrics.csv`.

Appendix A12: Model Family Winners

AutoGluon selection evidence

Every final feature-set/track run selected a `WeightedEnsemble_L2` as the top leaderboard model. The useful base signal is mostly tree-based; neural nets were considered but did not dominate.

Track	Feature set	Best family	Val. score	Fit sec.
Temporary	AG Expanded Dataset	WeightedEnsemble	0.2307	280
Temporary	AG Base Dataset	WeightedEnsemble	0.2203	83
Temporary	AG Latent Dataset (VAE)	WeightedEnsemble	0.1953	146
Temporary	AG Exp. Dataset + VAE	WeightedEnsemble	**0.2314**	203
Permanent	AG Expanded Dataset	WeightedEnsemble	0.2010	126
Permanent	AG Base Dataset	WeightedEnsemble	0.1929	229
Permanent	AG Latent Dataset (VAE)	WeightedEnsemble	0.1505	191
Permanent	AG Exp. Dataset + VAE	WeightedEnsemble	**0.2041**	291

Remark: Source: `AE-MODEL-SUITE-007_model_family_winners.csv`.

Appendix A13: Model Metric Caveats and OOS Robustness

Comparability

The raw comparator is the revised-dataset optional-library AutoGluon rerun from AE-VALIDATE. Complete local threshold metrics were later materialized under `03_Data_Output/6_ModelSuite/raw/derived_metrics`, so the presentation uses AP, AUC, Brier, and R@FPR1/3/5 consistently across feature sets.

- AP and fixed-FPR recall are emphasized because CSI positives are rare.
- AUC remains useful as a ranking diagnostic but can mask low-prevalence false-positive economics.
- AG Exp. Dataset + VAE is the main enhanced feature-set challenger for temporary CSI and permanent test/CV.
- AG Latent Dataset (VAE) is retained for permanent OOS robustness, not as a universal replacement.
- Economic superiority is not claimed from model metrics alone; index-construction slides test benchmark-relative performance.

Appendix A14: Final Index Grid Contract

Completed grid

- Models: AG Expanded Dataset, AG Base Dataset, AG Latent Dataset (VAE), and AG Exp. Dataset + VAE.
- Tracks: temporary CSI and permanent CSI.
- Universes: Total Market, Large Cap, Mid Cap, Small Cap.
- Thresholds: Youden, FPR1, FPR3, FPR5.

Portfolio rules

- Temporary CSI: 1-, 2-, 3-, and 5-year exclusion lockouts.
- Permanent CSI: absorbing permanent removal after first active signal.
- Costs: 0, 5, 10, and 20 bps on gross turnover.
- Turnover output is populated for every model–track–universe cell.

Presentation interpretation

The index section uses OOS economic performance. Model metrics identify useful prediction signals; index construction tests whether those signals improve benchmark-relative returns after thresholding, exclusions, turnover, and transaction costs.

Remark: Sources: `comparison/full_grid_manifest.csv`; `comparison/best_by_track_index_cost.csv`; `comparison/turnover_summary.csv`.

Appendix A15: 20 bps Winners With Benchmarks

Best strategy by track and universe at 20 bps

Track	Universe	Winner	Rule	Net ret.	Bench.	Alpha
Temp.	Total	AG Base Dataset	Youden / 3yr	13.70%	13.28%	+0.42pp
Temp.	Large	AG Base Dataset	Youden / 3yr	14.40%	14.25%	+0.15pp
Temp.	Mid	AG Base Dataset	FPR5 / 5yr	9.93%	9.42%	+0.51pp
Temp.	Small	AG Exp. Dataset + VAE	Youden / 3yr	8.11%	7.49%	+0.62pp
Perm.	Total	AG Exp. Dataset + VAE	FPR5 / perm.	13.54%	13.28%	+0.27pp
Perm.	Large	AG Exp. Dataset + VAE	FPR5 / perm.	14.48%	14.25%	+0.23pp
Perm.	Mid	AG Base Dataset	FPR5 / perm.	10.16%	9.42%	+0.74pp
Perm.	Small	AG Latent Dataset (VAE)	FPR3 / perm.	7.82%	7.49%	+0.32pp

VAE/non-raw variants beat raw in 5 of 8 track–index cases at 20 bps. The market-weighted benchmark is shown as the return reference and does not receive a strategy cost overlay. *Remark:* Source: `presentation_headline_tables.md`; final claim from `AE-INDEX-SUITE-009_Closeout_Report.md`.

Appendix A16: Error Decomposition and Index Source Paths

Base folders: `03_Data_Output/7_IndexConstructionValidation/nonraw_index_suite/`

- Headline winners: `final_tables/headline_winners_20bps.csv`
- Presentation tables: `final_tables/presentation_headline_tables.md`
- Transaction costs: `comparison/transaction_cost_impact.csv`; `final_tables/transaction_cost_robustness_summary.csv`
- Best-by-cost grid: `comparison/best_by_track_index_cost.csv`
- Threshold-family grid: `comparison/best_by_track_index_threshold_family.csv`
- Strategy performance fields: model-specific `index_performance_gross_and_net_by_tc.csv`
- Threshold families: `final_tables/threshold_family_summary_20bps.csv`
- Turnover: `comparison/turnover_summary.csv`; `final_tables/winner_turnover_summary_20bps.csv`
- Error-cost decomposition: selected AG Base Dataset, AG Exp. Dataset + VAE, and AG Latent Dataset (VAE) `error_cost_decomposition_by_crsp_universe.csv` files mapped exactly in `SLIDE_DATA_SOURCES.md`.

Source discipline

All claims in the index-result slides are sourced from the completed AE-INDEX-SUITE grid. Sensitivity-analysis results are intentionally not finalized in this section.

Appendix A17: Transaction-Cost Robustness

Top winner did not change between 0 and 20 bps

Track	Universe	Winner 0 bps	Winner 20 bps	Alpha 0	Alpha 20
Temp.	Total	AG Base Dataset	AG Base Dataset	+0.43pp	+0.42pp
Temp.	Large	AG Base Dataset	AG Base Dataset	+0.15pp	+0.15pp
Temp.	Mid	AG Base Dataset	AG Base Dataset	+0.51pp	+0.51pp
Temp.	Small	AG Exp. Dataset + VAE	AG Exp. Dataset + VAE	+0.64pp	+0.62pp
Perm.	Total	AG Exp. Dataset + VAE	AG Exp. Dataset + VAE	+0.27pp	+0.27pp
Perm.	Large	AG Exp. Dataset + VAE	AG Exp. Dataset + VAE	+0.23pp	+0.23pp
Perm.	Mid	AG Base Dataset	AG Base Dataset	+0.74pp	+0.74pp
Perm.	Small	AG Latent Dataset (VAE)	AG Latent Dataset (VAE)	+0.32pp	+0.32pp

Costs are applied at 0, 5, 10, and 20 bps. The 20 bps case is conservative enough for presentation, but not high enough to reverse the headline rankings. *Remark:* Source: `transaction_cost_robustness_summary.csv`.

Appendix A18: Threshold Family and Turnover Summary

Mean best alpha by threshold family

Track	Threshold	Mean alpha
Temp.	Youden	+0.35pp
Temp.	FPR3	+0.19pp
Temp.	FPR5	+0.17pp
Temp.	FPR1	+0.12pp
Perm.	FPR3	+0.35pp
Perm.	FPR5	+0.33pp
Perm.	FPR1	+0.14pp
Perm.	Youden	-0.20pp

Mean annualized turnover for winners

Universe	Temp.	Perm.
Total	15.3%	12.2%
Large	18.8%	17.2%
Mid	97.7%	97.0%
Small	77.2%	72.4%

Temporary CSI is strongest with Youden and longer finite lockouts; permanent CSI is strongest with FPR3/FPR5. Turnover is concentrated in Mid and Small Cap, where exclusions shift more of the investable universe. *Remark:* Sources: `threshold_family_summary_20bps.csv`; `winner_turnover_summary_20bps.csv`.

Appendix A19: Temporary CSI Sensitivity Detail

Temporary-CSI sensitivity only: selected configurations, main-run comparison, and limits

Role	Run ID	Pos.	Prev.	OOS AP	OOS AUC	11C TM alpha	Status
Composite	C090_M000_T012	12,761	7.02%	0.557	0.920	+0.169pp	complete
AP winner	C060_M000_T012	31,596	17.49%	0.566	0.864	+0.052pp	reused
11C winner	C090_M020_T018	6,035	3.34%	0.274	0.911	+0.226pp	complete
Baseline	C080_M020_T018	8,517	4.74%	0.308	0.896	+0.065pp	reused

Main comparison and stability

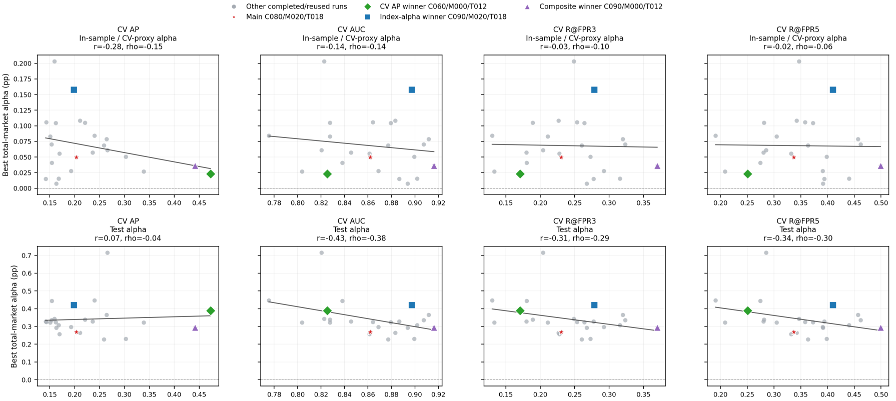
The accepted main run C080_M020_T018 is the continuity baseline, not the top sensitivity setting. Across 24 completed/reused C/M/T runs, benchmark-relative index alpha remains positive in all four universes; sensitivity changes magnitude more than direction.

Limits and transaction-cost overlay

The grid represents 27 temporary-CSI configurations: 24 are complete or reused, and three are `blocked_partial`: C080_M000_T012, C080_M000_T018, and C060_M020_T028. The sensitivity evidence does not support permanent-CSI claims. In the accepted-label temporary index grid, 0/5/10/20 bps overlays do not change the winner set.

Remark: Sources: local `sensitivity_cmt_model_summary.csv`; `sensitivity_cmt_index_summary.csv`; `temporary_index_transaction_cost_lines.csv`; `temporary_blocked_config_disclosure.csv`.

CV model metrics versus total-market index alpha across 24 completed temporary-CSI C/M/T runs



ML metrics are CV-only. Alpha uses the best non-benchmark total-market 11C strategy within each run and period. Insample is labelled as CV-proxy because the 11C source period is named insample, not CV.

- CV-only AP/AUC/R@FPR3/R@FPR5 do not map cleanly into total-market alpha; AP is near zero against test alpha.
- Index performance is the realized result after thresholding, timing, universe weights, and returns, not a generic ML metric.
- The 11C source period is named `insample`; this slide labels it as in-sample/CV-proxy alpha.

Remark: Sources: `slide48_cv_metrics_vs_cv_test_alpha.png`; local `sensitivity_cmt_model_summary.csv`; `combined_11c_performance.csv`; AE-PRES-QA-FIX-008 evidence.

Appendix A21: Future Robustness and Extensions

Not completed results

The items below are explicitly future validation or extensions. They should not be read as completed empirical findings in this deck.

Item	Purpose	Current status
Permanent-CSI sensitivity grid	Test label-threshold robustness for the permanent response track.	Planned / not completed
Minimum-volatility benchmark	Check whether CSI screens add value beyond a low-risk index comparator.	Planned / not completed
Quality/risk-scaling benchmark	Compare exclusions with quality or risk-scaled benchmark construction.	Planned / not completed
Active CSI-probability weighting	Use CSI probabilities continuously rather than thresholded exclusions.	Planned / not completed
Recovered bankrupt/insolvent diagnostic	Decompose remaining terminal-failure misses and recovery cases.	Planned / not completed

Remark: Source status is mapped in `SLIDE_DATA_SOURCES.md`; these are future-work notes, not result claims.

Appendix A22: Slide-to-Source Audit Map

Audit artifact

The complete slide-to-source map is stored beside this June source deck:
`06_Presentations/02_FinalPresentation/Necessary/FinalPresentation_June/SLIDE_DATA_SOURCES.md`.

- Dataset and label slides map to AE-PANEL evidence and descriptive-statistics CSVs.
- Model slides map to AE-MODEL-SUITE comparison files and threshold metrics.
- Index slides map to AE-INDEX-SUITE final-table and comparison outputs.
- Sensitivity slides map to local AE-SENS manifests and comparison artifacts.
- Conceptual or future-work slides are marked as such rather than linked to empirical result files.

Remark: This appendix frame is a navigation pointer; the source map contains per-frame paths and source type.

Bibliography I

Temporary CSI Test-Set Index Results at 10 bps

2016–2019 test-set benchmark and best temporary-CSI strategy by universe

Universe	Strategy	Geo ret.	Ann. SD	Sharpe Ratio	Max DD	ES 2.5%	Delta pp
Total	Benchmark	13.75%	12.05%	0.889	-14.62%	-8.44%	0.00pp
Total	AG Exp. Dataset + VAE; Youden/3yr	14.15%	11.65%	0.946	-13.78%	-8.14%	+0.41pp
Large	Benchmark	14.17%	11.60%	0.951	-13.82%	-8.02%	0.00pp
Large	AG Exp. Dataset + VAE; Youden/3yr	14.41%	11.40%	0.984	-13.33%	-7.86%	+0.24pp
Mid	Benchmark	13.73%	12.85%	0.840	-15.32%	-8.69%	0.00pp
Mid	AG Expanded Dataset; Youden/5yr	14.06%	12.15%	0.905	-14.41%	-8.34%	+0.33pp
Small	Benchmark	11.37%	15.55%	0.582	-20.09%	-10.81%	0.00pp
Small	AG Exp. Dataset + VAE; Youden/2yr	12.26%	14.47%	0.670	-18.37%	-10.13%	+0.89pp

Test-set check

These are isolated 2016–2019 test rows at 10 bps, not the 2020+ OOS index construction. The validation package contains `period=test` rows only and reports zero OOS rows.

Remark: Source: `03_Data_Output/9_TestIndexConstruction/AE-FP-DIAG-006_test_index_performance_gross_and_net_by_tc.csv`; validation: `AE-FP-DIAG-006_validation_checks.csv`.

Temporary CSI Test-Set Diagnostic and Active Contribution

Selected test-set winners at 10 bps; contribution rows use main-suite `period=test` attribution

Universe	Strategy	TP gain	FP cost	Reweight	TC	Geo adj.	Alpha
Total	AG +VAE, Youden 3yr	+0.08pp	-0.38pp	+0.67pp	-0.01pp	+0.04pp	+0.40pp
Large	AG +VAE, Youden 3yr	+0.00pp	-0.19pp	+0.41pp	-0.01pp	+0.02pp	+0.23pp
Mid	AG Expanded, Youden 5yr	+0.01pp	-0.67pp	+1.07pp	-0.09pp	-0.08pp	+0.24pp
Small	AG +VAE, Youden 2yr	+0.31pp	-1.19pp	+1.42pp	-0.07pp	+0.34pp	+0.82pp

Attribution caveat

The result table above is from isolated test-output files. This diagnostic view is not a standalone AE-FP-DIAG-006 test diagnostic build; it uses reconciled main-suite `period=test` attribution rows for the same temporary-CSI strategies.

In the test window, gains mostly come from retained-stock reweighting and geometric effects after false-positive costs. TP exclusion helps most visibly in Small Cap, but the slide should not be read as causal event avoidance. *Remark:* Sources: `AE-ATTRIB-001_config_level_attribution.csv`; model-specific `error_cost_decomposition_by_crsp_universe.csv`; `transaction_cost_bps=10`, `period=test`; reconciliation checks in `AE-ATTRIB-001_reconciliation_checks.csv`.